WORKED SOLUTION

STEP III 1993, Question 12

Mechanics • Elastic Strings and Energy

STEP III 1993, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

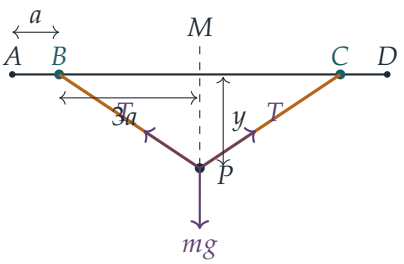
step.maths.org/questions/93-s3-q12

The key idea

Because the pegs are smooth and the particle sits at the midpoint, the string splits into two identical halves, each of natural length a, with the same tension all along each half. Each half runs from a fixed end, over a peg, down (or up) to P, so its stretched length depends only on the distance BP. Once we write the extension in terms of the height of P, equilibrium gives k and conservation of energy links Q to R.

Solution

We model P as a particle, the string as light and elastic, and the pegs as smooth. By symmetry P moves on the vertical line through the midpoint M of BC. Let y be the depth of P below M; it is negative when P is above BC.



Finding k

The half $A \to B \to P$ has length $a + BP = a + \sqrt{9a^2 + y^2}$ and natural length a, so its extension is

$$x = \sqrt{9a^2 + y^2}.$$

By Hooke's law the tension in each half is

$$T = \frac{kmg\,x}{a}.$$

If θ is the angle BP makes with the horizontal, then $\sin\theta = |y|/x$. Resolving vertically at P gives

$$2T\sin\theta = mg \quad\Longrightarrow\quad 2\cdot\frac{kmg\,x}{a}\cdot\frac{y}{x} = mg \quad\Longrightarrow\quad \frac{2ky}{a} = 1.$$

The square root has cancelled, so the ugly value $BP = \frac{\sqrt{145}}{4}a$ never matters. With $y = a/4$ we get $k/2 = 1$, so

$$k = 2.$$

Energy from Q to R

Each half stores elastic energy $\dfrac{kmg\,x^2}{2a}$, so with $k = 2$ the two halves together store

$$E(y) = 2\cdot\frac{2mg\,x^2}{2a} = \frac{2mg\,(9a^2+y^2)}{a}.$$

Ask yourself Why does P stop at R, and why is it fine that P ends up above the pegs?

P is released from rest at Q, and R is the highest point it rises to, so it is momentarily at rest there too. Above BC the halves are still stretched (each still has $x \geq 3a > 0$), so the same formula for E applies. The string never goes slack.

At Q, $y = pa$; at R, $y = -3a$. The particle rises $(p+3)a$. Since the pegs are smooth and there is no other loss, the kinetic energy is zero at both ends, so

$$\begin{aligned} E(pa) &= E(-3a) + mg(p+3)a \\ 2mga(9+p^2) &= 2mga(9+9) + mga(p+3) \\ 18 + 2p^2 &= 36 + p + 3, \end{aligned}$$

which gives

$$2p^2 - p - 21 = 0, \qquad \text{that is,} \qquad (2p+6)(p-\tfrac{7}{2}) = 0.$$

Since $p > 0$, we get $p = \frac{7}{2}$.

This does *not* agree with the printed $2p^2 - p - 17 = 0$. We have checked the model carefully (see Checks), including a direct numerical simulation, and -21 is what the stated data give. The clean root $p = \frac{7}{2}$ suggests that 21 was the intended constant and that 17 is a misprint. The final part does not depend on which constant is right.

To confirm that P really does stop for the first time at R, write $y = ta$. The kinetic energy at depth ta is

$$E(pa) - E(ta) - mg(p-t)a = mga\,(p-t)(2p+2t-1).$$

With $p = \frac{7}{2}$ this is $mga\,(\frac{7}{2}-t)(2t+6)$. It is positive for $-3 < t < \frac{7}{2}$ and first vanishes at $t = -3$, which is R.

Comparing the tensions

The tension is $T = 2mg\,x/a$, which increases with x, so we need to show that x is smaller at R than at Q. The energy equation does this without using the value of p:

$$E(Q) - E(R) = mg(p+3)a > 0.$$

The elastic energy is $\dfrac{2mg\,x^2}{a}$, so $x_R^2 < x_Q^2$, giving $x_R < x_Q$ and hence $T_R < T_Q$. In words, the particle gains potential energy on the way up, and that energy can only come out of the string, so the string must be less stretched at R.

Explicitly, with $p = \frac{7}{2}$, $x_Q = \frac{\sqrt{85}}{2}a$ and $x_R = 3\sqrt{2}\,a$, so

$$T_Q = \sqrt{85}\,mg \approx 9.22\,mg, \qquad T_R = 6\sqrt{2}\,mg \approx 8.49\,mg.$$

Checks

- *SHM check.* The upward force on P is $2T \cdot \dfrac{y}{x} = \dfrac{4mg\,y}{a}$, which is linear in y, above BC as well as below. So P moves in simple harmonic motion about $y = a/4$ (the equilibrium found above). Released from rest at $y = \frac{7}{2}a$, its amplitude is $\frac{13}{4}a$, so its highest point is $y = \frac{1}{4}a - \frac{13}{4}a = -3a$. This is exactly R, which confirms $p = \frac{7}{2}$ independently.
- A numerical integration of the equation of motion with $a = 1$, $k = 2$, $p = 3.5$ reaches a highest point of $y = -3.0000$.
- The printed equation would give $p = \frac{1+\sqrt{137}}{4} \approx 3.18$, and the energy balance then fails by $4mga$.

Common slips

- Leaving out the segment AB when computing the length of each half. The half is $AB + BP = a + BP$ long, and its extension is BP only because AB equals its natural length a.
- Treating the whole string as one spring with tension $kmg \cdot (\text{total extension})/(2a)$. This does give the same answer, but only if the extension is the *total* one, $2x$.
- Forgetting the elastic energy is still there when P is above BC. The string is still stretched there.

Beyond STEP

The vertical restoring force came out exactly proportional to displacement, even though the string geometry is nonlinear. This is because the vertical component of a tension proportional to length is proportional to the vertical separation, the same reason a

zero-natural-length spring makes an ideal suspension (the principle behind the LaCoste long-period seismometer). Potential energy that is quadratic in the coordinates leads to linear equations of motion, and this is the starting point of the theory of small oscillations and normal modes. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats springs, energy and oscillations in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.